Ethan McCormick, Ph.D.

Curriculum Vitae

✉ *ethan.mccormick@rutgers.edu*

Education

2020 **Ph.D. in Psychology & Neuroscience**, *Concentration in Developmental and Quantitative*, *University of North Carolina at Chapel Hill*
Advisors: Dr. Eva Telzer, Dr. Patrick Curran, Dr. Kathleen Gates
Dissertation Title: "Neural Network Plasticity in Response to Experience: Contributions of Learning and Development".

2016 **M.A. in Psychology**, *University of Illinois, Urbana-Champaign*.

2012 **B.S in Biochemistry**, *University of Arkansas (with honors)*.

Employment Experience

2020-present **Postdoctoral Fellow**, Cole Neurocognition Laboratory, *Center for Molecular and Behavioral Neuroscience*.
Rutgers University, Newark

2020-present **Consultant**, Curran-Bauer Analytics.
Classes: Multi-level Modeling, Structural Equation Modeling, Longitudinal Structural Equation Modeling, Network Analysis, Mixture Modeling

Competitive Awards and Honors

2019 Stanford PRISM Fellow

2019 ABCD Summer Workshop Travel Award ($500)

2018 Dashiell Departmental Travel Award, UNC ($1000)

2017-2018 Dashiell Dissertation Startup Award, UNC ($1000)

2017-2018 Ernest C. Davenport Award for Outstanding Research by a Student Who Enhances Diversity, UNC ($200)

2015 Social Affective Neuroscience Society Travel Grant ($250)

2014-2015 Rhymer's Fellowship, UIUC

Grant Activity

2020 **Graduate Student Consultant**, *Neurobiological susceptibility to peer influence and drug use in adolescence*,
NIDA R01 DA051127,
Mitch Prinstein, Kristen Lindquist, & Eva Telzer, PIs.
University of North Carolina at Chapel Hill

2020 **Co-Investigator**, *Modeling Developmental Change in the ABCD Study: Longitudinal Analyses for Clinical Outcomes*,
NIMH: 1R25MH125545-01,
Kathryn Mills, PI.
University of Oregon

Software

contributor **gimme (R-package)**, *Group Iterative Multiple Model Estimation*, [Github], [CRAN].

creator **gimmereg (R-package)**, *Directed functional connectivity search algorithm using regularized regression*, [Github], [CRAN].

Preprints

1. **McCormick, E.M.** (*preprint*). Multi-Level Multi-Growth Models: New opportunities for addressing developmental theory using longitudinal designs. https://doi.org/10.1101/2020.10.21.349274

Publications

25. **McCormick, E.M.**, Peters, S., Crone, E.A., & Telzer, E.H. (2021). Longitudinal Network Re-organization Across Learning and Development. *NeuroImage*. 117784.
24. Duell, N., van Hoorn, J., **McCormick, E.M.**, Prinstein, M.J., & Telzer, E.H. (*in press*). Hormonal and neural correlates of prosocial conformity in adolescents. *Developmental Cognitive Neuroscience*. Accepted January 2021.
23. Kwon, S-J., Do, K.T., **McCormick, E.M.**, & Telzer, E.H. (*in press*). Neural correlates of conflicting social influence on adolescent risk-taking. *Journal of Research on Adolescence*. Accepted September 2020.
22. Do, K.T, **McCormick, E.M.**, & Telzer, E.H. (2020). Neural sensitivity to conflicting attitudes supports greater conformity toward positive over negative influence in early adolescence. *Developmental Cognitive Neuroscience*, 45. 100837.
21. Van Hoorn, J., **McCormick, E.M.**, Perino, M.T., Rogers, C.R., & Telzer, E.H. (2020). Differential behavioral and neural profiles in high-risk youth with conduct problems during risky decision-making. *Journal of Research on Adolescence, 30*(3), 599-615.
20. Chen, X., **McCormick, E.M.**, Ravindran, N., Telzer, E.H., & McElwain, N.L. (2020). Maternal emotion socialization in early childhood predicts adolescents' amygdala-vmPFC functional connectivity to emotion faces. *Developmental Psychology, 56*(3), 503-515.
19. Perino, M.T., Guassi Moreira, J., **McCormick, E.M.**, & Telzer, E.H. (2019). Apples to apples? Neural correlates of emotion regulation differences between high and low risk adolescents. *Social Cognitive and Affective Neuroscience, 14*(8), 827-836.
18. Kwon, S-J., Ivory, S.L., **McCormick, E.M.**, & Telzer, E.H. (2019). Behavioral and neural dysregulation to social reward and links to internalizing symptoms in adolescents. *Frontiers in Behavioral Neuroscience, 13*, 158.
17. **McCormick, E.M.**, McElwain, N.L., & Telzer, E.H. (2019). Alterations in adolescent dopaminergic systems as a function of early mother-toddler attachment: a prospective longitudinal examination. *International Journal of Developmental Neuroscience, 78*, 122-129.

16. Do, K.T, **McCormick, E.M.**, & Telzer, E.H. (2019). The neural development of prosocial behavior from childhood to adolescence. *Social Cognitive and Affective Neuroscience, 14*, 129-139.
15. **McCormick, E.M.**, Gates, K.M., & Telzer, E.H. (2019). Model-based network discovery of developmental and performance-related differences during risky decision-making. *NeuroImage, 188*, 456-464.
14. **McCormick, E.M.**, & Telzer, E.H. (2018b). Differential contributions of default mode network stability and deactivation to adolescent task engagement. *Scientific Reports, 8(1)*, 18049.
13. Van Hoorn, J., **McCormick, E.M.**, Rogers, C.R., Ivory, S.L., & Telzer, E.H. (2018). Differential effects of parent and peer presence on neural correlates of risk taking in adolescence. *Social Cognitive and Affective Neuroscience, 13*(9), 945-955.
12. **McCormick, E.M.**, Van Hoorn, J., Cohen, J.R., & Telzer, E.H. (2018). Functional Connectivity in the social brain across childhood and adolescence. *Social Cognitive and Affective Neuroscience, 13*(8), 819-830.
11. Rogers, C.R., **McCormick, E.M.**, Van Hoorn, J., & Telzer, E.H. (2018). Neural correlates of sibling closeness and association with externalizing behavior in adolescence. *Social Cognitive and Affective Neuroscience, 13*(9), 977-988.
10. Muscatell, K.A., **McCormick, E.M.**, & Telzer, E.H. (2018). Subjective social status and neural processing of race in Mexican American adolescents. *Development and Psychopathology*, 1-12.
9. Van Hoorn, J., **McCormick, E.M.**, & Telzer, E.H. (2018). Moderate social sensitivity in a risky context supports adaptive decision-making in adolescence: Evidence from brain and behavior. *Social Cognitive and Affective Neuroscience, 13*(5), 546-556.
8. Telzer, E.H., **McCormick, E.M.**, Peters, S., Cosme, D., Pfiefer, J.H., & van Duijvenvoorde, A.C.K. (2018). Methodological considerations for developmental longitudinal fMRI research. *Developmental Cognitive Neuroscience, 33*, 149-160.
7. **McCormick, E.M.**, Perino, M.T., & Telzer, E.H. (2018). Not just social sensitivity: Selective adolescent suppression of social feedback during risk taking. *Developmental Cognitive Neuroscience, 30*, 134-141.
6. **McCormick, E.M.**, & Telzer, E.H. (2018a). Not doomed to repeat: Enhanced neural tracking of errors promotes adaptive task performance during adolescence. *Journal of Cognitive Neuroscience. 30*(3), 281-289.
5. Qu, Y., Pomerantz, E.M., **McCormick, E.M.**, & Telzer, E.H. (2018). Youth's conceptions of adolescence predict longitudinal changes in prefrontal cortex activation and risk taking. *Child Development, 89*(3), 773-783.
4. **McCormick, E.M.**, Qu, Y., & Telzer, E.H. (2017). Activation in context: Differential conclusions drawn from cross-sectional and longitudinal analyses of adolescents' cognitive control-related neural activity. *Frontiers in Human Neuroscience, 11*, 141.
3. **McCormick, E.M.**, & Telzer, E.H. (2017b). Failure to retreat: Blunted sensitivity to negative feedback supports risky behavior in adolescents. *NeuroImage, 147*, 381-389.
2. **McCormick, E.M.**, & Telzer, E.H (2017a). Adaptive adolescent flexibility: Neurodevelopment of decision-making and learning in a risky context. *Journal of Cognitive Neuroscience, 29*, 413-423.

1. **McCormick, E.M.**, Qu, Y., & Telzer, E.H. (2016). Adolescent neurodevelopment of cognitive control and risk-taking in negative family contexts. *NeuroImage, 124*, 989-996.

Manuscripts Under Review

2. **McCormick, E.M.** (*under review*). Multi-Level Multi-Growth Models: New opportunities for addressing developmental theory using longitudinal designs. *Developmental Cognitive Neuroscience*

1. Do, K.T, **McCormick, E.M.**, Lindquist, K.A., Prinstein, M.J., & Telzer, E.H. (*revise and resubmit*). Differential susceptibility to peer influence in adolescence: The role of intrinsic functional connectivity within social-affective brain networks. *Developmental Science*

Manuscripts in Preparation

5. **McCormick, E.M.**, Arnemann, K., Ito, T., & Cole, M.W. (*in progress*). Estimating latent functional connectivity underlying multiple brain states.

4. Curran, P.J., Strauss, C., **McCormick, E.M.**, & McGinley, J.S. (*in progress*). A multivariate growth curve model for three-level data. In H. Cooper, P.M. Camic, D.L. Long, A.T. Panter, D. Rindskopf, & K.J. Sher (Eds.), APA Handbooks in Psychology®. APA Handbook of Research Methods in Psychology, Vol. 3. Data Analysis and Research Publication, Second Edition. American Psychological Association.

3. Duell, N., **McCormick, E.M.**, Perino, M.T., & Telzer, E.H. (*in progress*). Maturing Immaturely: Mismatches between neural sensitivity to risk and negative social information in adolescents with severe conduct problems.

2. **McCormick, E.M.**, & Telzer, E.H. (*in progress*). Feedback processing and learning: Extracting information from an uncertain environment.

1. **McCormick, E.M.**, Gates, K.M., & Ye, A. (*in progress*). Directed connectivity search through regularized regression using a group-information heuristic.

Invited Talk Presentations

9. **McCormick, E.M.**, (December 2020). Multi-level, multi-growth models: new opportunities for addressing developmental theory. Talk presented at the University of Melbourne, AU (remotely).

8. **McCormick, E.M.**, (November 2020). Multi-level, multi-growth models: new opportunities for addressing developmental theory. Talks presented at Cambridge University, UK and the University of Oregon, OR (remotely).

7. **McCormick, E.M.**, (August 2019). Constructing model-based networks. Talk presented at the ABCD Workshop on Brain Development and Mental Health, Portland, OR.

6. **McCormick, E.M.**, (November 2018). Measuring Neural Change: Short and long-term plasticity in neural networks. Talk presented at the University of Pennsylvania, PA.

5. **McCormick, E.M.**, (May 2018). Plasticity during adolescence: neural systems which support flexible behavior. Talk presented at the University of Leiden, Leiden, NL.

4. **McCormick, E.M.**, (March 2018). Neural Plasticity: Change in Response to Experience and Development. Talk presented at the University of North Carolina, Chapel Hill Developmental Lunch, Chapel Hill, NC.

3. **McCormick, E.M.**, (February 2017). Neural Markers of Adaptive Flexibility in Adolescence. Talk presented at the University of North Carolina, Chapel Hill Developmental Lunch, Chapel Hill, NC.
2. **McCormick, E.M.**, (April 2016). fMRI Study of Cognitive Control in Teens âĂŞ- Implications of cross-sectional vs. longitudinal analyses. Talk presented at the Modeling Longitudinal Data Meeting, Eugene, OR.
1. **McCormick, E.M.**, (April 2015). Longitudinal Links between Neural Development and Behavior in Early Adolescence. Talk presented at the University of Illinois, Urbana-Champaign Developmental Brownbag, Champaign, IL.

Conference Talk Presentations

9. McElwain, N.L., Chen, X., Ravindran, N., **McCormick, E.M.**, & Telzer, E.H. (March 2019). Maternal Sensitivity during Toddlerhood Predicts Amygdala-PFC Functional Connectivity During Early Adolescence. Talk presented at the 2019 Biennial Meeting of the Society for Research in Child Development, Baltimore, MD.
8. Telzer., E.H., Ivory, S., & **McCormick, E.M.** (March 2019). Altered neural connectivity and cognitive control to peer faces: Links to internalizing symptoms in adolescence. Talk presented at the 2019 Biennial Meeting of the Society for Research in Child Development, Baltimore, MD.
7. Telzer, E.H., **McCormick, E.M.**, McElwain, N.L. & Qu. Y. (March 2019). Family relationship quality and adolescent neural processing of risk taking. Talk presented at the 2019 Biennial Meeting of the Society for Research in Child Development, Baltimore, MD.
6. Van Hoorn, J., **McCormick, E.M.**, Rogers, C.R., & Telzer, E.H. (April 2018). Risk taking and peer effects in high-risk youth with externalizing behavior: Perspectives from brain and behavior. Talk presented at the 2018 Society for Research on Adolescence Biennial Meeting, Minneapolis, MN.
5. Rogers, C.R., **McCormick, E.M.**, van Hoorn, J., & Telzer, E.H. (April 2018). Siblings and the teenage brain: sibling closeness and birth order modulate adolescent neural activity during safe decision-making. Talk presented at 2018 Society for Research on Adolescence Biennial Meeting, Minneapolis, MN.
4. Rogers, C.R., **McCormick, E.M.**, van Hoorn, J., & Telzer, E.H. (September 2017). No, don't do it!" Neural correlates of sibling closeness during risky decision-making. Talk presented at the 2017 Flux Congress Annual Meeting, Portland, OR.
3. **McCormick, E.M.**, & Telzer, E.H. (April 2017). Adaptive adolescent flexibility: neurodevelopment of decision-making and learning in a risky context. Talk presented at the Biennial Meeting of the Society for Research in Child Development, Austin, TX.
2. **McCormick, E.M.**, & Telzer, E.H. (March 2016). Activation in context: Cross-Sectional and longitudinal analyses of adolescents' cognitive-control related neural activity. Talk presented at the Society for Research on Adolescence Biennial Meeting, Baltimore, MD.
1. Telzer, E.H., & **McCormick, E.M.** (November, 2015). Differential conclusions drawn for cross-sectional longitudinal analyses of adolescents' cognitive-control related neural activity: best practices fore examining brain behavior associations? Talk presented at the Conference on Longitudinal fMRI Analysis, Stockholm, SE.

Conference Poster Presentations

21. **McCormick, E.M.**, Telzer, E.H, & Gates, K.M. (August 2019). Reliability in clustering solutions derived from resting state fMRI: Insights from the Human Connectome Project. Poster presented at the 2019 Flux Congress Annual Meeting, New York City, NY.
20. Duell, N., Van Hoorn, J., **McCormick, E.M.**, & Telzer, E.H (August 2019). The culture of socioeconomic status and social reward processing in adolescence. Poster presented at the 2019 Flux Congress Annual Meeting, New York City, NY.
19. Jorgensen, N.A., **McCormick, E.M.**, Lindquist, K.A., Prinstein, M.J., & Telzer, E.H (August 2019). The culture of socioeconomic status and social reward processing in adolescence. Poster presented at the 2019 Flux Congress Annual Meeting, New York City, NY.
18. Kwon, S-J., Do, K.T., **McCormick, E.M.**, & Telzer, E.H (August 2019). Neural correlates of conflicting social influences on adolescent risk-taking. Poster presented at the 2019 Flux Congress Annual Meeting, New York City, NY.
17. Ravindran, N., McElwain, N.L., **McCormick, E.M.**, & Telzer, E.H (March 2019). Warmth and negativity in mother-adolescent relationships: Associations with adolescents' neural responses to angry faces. Poster presented at the 2019 Biennial Meeting of the Society for Research in Child Development, Baltimore, MD.
16. Chen, X., **McCormick, E.M.**, McElwain, N.L., & Telzer, E.H (March 2019). Maternal emotion talk in early childhood predicts adolescents' neural activity when labeling facial emotion. Poster presented at the 2019 Biennial Meeting of the Society for Research in Child Development, Baltimore, MD.
15. **McCormick, E.M.**, & Telzer, E.H. (September 2018). Model-based network discovery of developmental and performance-related differences during risky decision-making. Poster presented at the 2018 Flux Congress Annual Meeting, Berlin, DE.
14. **McCormick, E.M.**, & Telzer, E.H. (May 2018). Functional dynamics of the social brain: Network insights from the Human Connectome Project. Poster presented at the 2018 Flux Society Satellite Conference, Chapel Hill, North Carolina.
13. Chen, X., **McCormick, E.M.**, McElwain, N.L., & Telzer, E.H. (May 2018). Maternal mental-state talk in early childhood predicts adolescents' neural activity to ambiguous facial emotions. Poster presented at the 2018 Flux Society Satellite Conference, Chapel Hill, North Carolina.
12. Qu, Y., Pomerantz, E.M., **McCormick, E.M.**, & Telzer, E.H. (April 2018). Youth's conceptions of adolescence predict longituidnal changes in prefrontal cortex activation and risk taking during adolescence. Poster presented at the 2018 Social for Research on Adolescence Biennial Meeting, Minneapolis, Minnesota.
11. Do, K.T., **McCormick, E.M.**, & Telzer, E.H. (April 2018). Is blood thicker than water? How conflicting parent and peer attitudes influence the neural correlates of adolescent conformity. Poster presented at the 2018 Social for Research on Adolescence Biennial Meeting, Minneapolis, Minnesota.
10. **McCormick, E.M.**, van Hoorn, J., & Telzer, E.H. (March 2017). Functional network architecture of the social brain during childhood and adolescence. Poster presented at the 2017 Flux Congress Annual Meeting, Portland, OR.
9. Ivory, S.I., **McCormick, E.M.**, & Telzer, E.H. (March 2017). Can't fight this feeling: The impact of emotional faces on adolescents' cognitive control. Poster presented at the 2017 Meeting of the Social and Affective Neuroscience Society, Los Angeles, CA.

8. van Hoorn, J., **McCormick, E.M.**, & Telzer, E.H. (March 2017). Social learning and adaptive risk-taking in adolescence: Evidence from brain and behavior. Poster presented at the 2017 Meeting of the Social and Affective Neuroscience Society, Los Angeles, CA.
7. **McCormick, E.M.**, & Telzer, E.H. (March 2017). Not just social sensitivity: Selective adolescent suppression of social information during risk taking. Poster accepted to the 2017 Meeting of the Social and Affective Neuroscience Society, Los Angeles, CA.
6. **McCormick, E.M.**, & Telzer, E.H. (September 2016). Two roads diverge: Context-specific outcomes associated with decreased neural sensitivity to negative feedback during adolescence. Poster presented at the Fourth Annual Meeting of the Flux Congress, St. Louis, MO
5. **McCormick, E.M.**, & Telzer, E.H. (April 2016). Adaptive adolescent flexibility: Decision-making in a risky context. Poster presented at the 2016 Meeting of the Social and Affective Neuroscience Society, New York, NY.
4. Do, K.T., **McCormick, E.M.**, & Telzer, E.H. (April 2016). Prosocial and social brain development from childhood to adolescence. Poster presented at the 2016 Meeting of the Social and Affective Neuroscience Society, New York, NY.
3. **McCormick, E.M.**, & Telzer, E.H. (September 2015). Longitudinal links between negative family relationships and adolescent cognitive control-related neural processing. Poster presented at the Third Annual Meeting of the Flux Congress, Leiden, NL.
2. **McCormick, E.M.**, & Telzer, E.H. (September 2015). Longitudinal links between negative family relationships and adolescent cognitive control-related neural processing. Poster presented at the Satellite Meeting for the Society for Research in Child Development, Leiden, NL.
1. **McCormick, E.M.**, & Telzer, E.H. (April 2015). Necessity of longitudinal analyses for understanding the neural development of cognitive control. Poster presented at the 2015 Meeting of the Social and Affective Neuroscience Society, Boston, MA.

Methodological and Technical Skills

Programming	MATLAB, Python, R, Visual Basics, Bash
Analysis Techniques	ANOVA, Regression, Mediation, Moderation, Multi-Level Modeling, Structural Equation Modeling, Latent Curve Modeling, Computational Modeling, Neural Network Analyses, Data Simulation, CART Analysis
Statistical Software	R, SAS, MPlus, Python, MATLAB, HLM, SPSS

Institutional Service

2019	Neuroscience Club Graduate School Panel, UNC
2018	fMRI Methodology & Coding Summer Seminar, UNC
2018	NeuroGrads Organizer, UNC
2017-2018	Diversity Admissions Committee, UNC
2017-2018	Developmental Seminar Planning Committee, UNC
2017	fMRI Methodology & Coding Summer Seminar, UNC
2016	fMRI Methodology & Coding Summer Seminar, UNC

2015-2016 Graduate Student Recruitment Organizer, UIUC
2015 fMRI Methodology Summer Seminar, UNC

Professional Affiliations

Member Flux Society for Developmental Cognitive Neuroscience
Member Social and Affective Neuroscience Society
Member Society for Research on Child Development

Ad Hoc Reviewer

Biological Psychiatry (under the direction of Dr. Eva Telzer)
Cognitive, Behavioral, and Affective Neuroscience (under the direction of Dr. Eva Telzer)
Developmental Cognitive Neuroscience (under the direction of Dr. Eva Telzer)
Frontiers in Human Neuroscience (under the direction of Dr. Eva Telzer)
Journal of Neuroscience (under the direction of Dr. Eva Telzer)
NeuroImage (under the direction of Dr. Eva Telzer)
Social Cognitive and Affective Neuroscience (under the direction of Dr. Eva Telzer)
Trends in Cognitive Science (under the direction of Dr. Eva Telzer)
Nature Neuroscience (under the direction of Dr. Eva Telzer)
Social Cognitive and Affective Neuroscience (independent)
NeuroImage (independent)
Cognitive, Affective, & Behavioral Neuroscience (independent)
Developmental Cognitive Neuroscience (independent)

Teaching Experience

Summer 2020 **Instructor**, *Introduction to Statistics*,
University of North Carolina at Chapel Hill.

Fall 2019 **Graduate Student Consultant**, *Clinical Research - Design, Analyze, Disseminate*,
Instructor: Dr. Desirée Griffin, University of North Carolina at Chapel Hill.

Fall 2019 **Teaching Assistant**, *Introduction to Neuroscience*,
Instructor: Dr. Sabrina Robertson, University of North Carolina at Chapel Hill.

Spring 2019 **Teaching Assistant**, *Introduction to Statistics*,
Instructor: Dr. Viji Sathy, University of North Carolina at Chapel Hill.

Fall 2017 **Teaching Assistant**, *Child Development*,
Instructor: Dr. Jean-Louis Gariepy, University of North Carolina at Chapel Hill.

Fall 2016 **Teaching Assistant**, *Introduction to Neuroscience*,
Instructor: Dr. Marsha Penner, University of North Carolina at Chapel Hill.

Spring 2016 **Teaching Assistant**, *Psychology of the Infant*,
Instructor: Dr. Lisa Travis, University of Illinois.

Fall 2016 **Teaching Assistant**, *Developmental Psychopathology*,
Instructor: Dr. Karen Rudolph, University of Illinois.

Spring 2014 **Undergraduate Teaching Assistant**, *Behavioral Neuroscience*,
Instructor: Dr. Nathan Parks, University of Arkansas.

Spring 2009-Fall 2013	**Undergraduate Supplemental Instructor**, *Organic Chemistry I & II*, Supervisor: Anne Raines, University of Arkansas.